

KITABU QUEST

S3

ICT



Cover scene

learners using computers safely with a mountain gorilla mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

3

Title Page

KITABU QUEST S3 ICT

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S3 learners study ICT one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Practise safe, useful digital skills and explain each step clearly.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Application and utility software installation
2. Charts and advanced objects
3. Arc GIS
4. Charts and objects in spreadsheet
5. Presentation
6. Network security
7. Platform game programming

As you finish each topic, return to the success criteria and correct one answer before moving on.

Application and utility software installation: Lesson Opener

Learning focus: Application and utility software installation

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Application and utility software installation.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- understand the functions of operating system, install and use application and utility software

Inquiry questions:

- How can application and utility software installation help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Application and utility software installation: Learn

Application and utility software installation is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Application and utility software installation
- Application
- utility
- software
- installation
- idea
- skill
- example

Application and utility software installation: Worked Example

Worked example for Application and utility software installation: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called application-and-utility-software-installation-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Application and utility software installation: Vocabulary And Study Skill

Use these words and study moves before you practise Application and utility software installation. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Application and utility software installation
- Application
- utility
- software
- installation
- idea
- skill
- example

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Application and utility software installation without a clear example, method, or evidence.

Application and utility software installation: Guided Activity

Guided digital activity for Application and utility software installation:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Application and utility software installation: Practice And Correction

Digital practice for Application and utility software installation:

1. Complete a small digital-skills task for Application and utility software installation: understand the functions of operating system, install and use application and utility software. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Application and utility software installation: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Application and utility software installation.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Application and utility software installation: Outcome Check

Outcome check for Application and utility software installation: Understand the functions of operating system, install and use application and utility software.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Charts and advanced objects: Lesson Opener

Learning focus: Charts and advanced objects

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Charts and advanced objects.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- present data graphically and use different objects in a document

Inquiry questions:

- How can charts and advanced objects help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Charts and advanced objects: Learn

Charts and advanced objects is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Charts and advanced objects
- Charts
- advanced
- objects
- idea
- skill
- example
- practice

Charts and advanced objects: Worked Example

Worked example for Charts and advanced objects: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called charts-and-advanced-objects-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Charts and advanced objects: Vocabulary And Study Skill

Use these words and study moves before you practise Charts and advanced objects. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Charts and advanced objects
- Charts
- advanced
- objects
- idea
- skill
- example
- practice

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Charts and advanced objects without a clear example, method, or evidence.

Charts and advanced objects: Guided Activity

Guided digital activity for Charts and advanced objects:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Charts and advanced objects: Practice And Correction

Digital practice for Charts and advanced objects:

1. Complete a small digital-skills task for Charts and advanced objects: present data graphically and use different objects in a document. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Charts and advanced objects: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Charts and advanced objects.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Charts and advanced objects: Outcome Check

Outcome check for Charts and advanced objects: Present data graphically and use different objects in a document.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Arc GIS: Lesson Opener

Learning focus: Arc GIS

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Arc GIS.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- query a map using selections (attribute and location). Create a map layout and insert map elements. Use advanced symbology and labeling. Print, export and send a map

Inquiry questions:

- How can arcgis help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Arc GIS: Learn

Arc GIS is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Arc GIS
- idea
- skill
- example
- practice
- correction

Arc GIS: Worked Example

Worked example for Arc GIS: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called arc-gis-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Arc GIS: Vocabulary And Study Skill

Use these words and study moves before you practise Arc GIS. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Arc GIS
- idea
- skill
- example

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Arc GIS without a clear example, method, or evidence.

Arc GIS: Guided Activity

Guided digital activity for Arc GIS:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Arc GIS: Practice And Correction

Digital practice for Arc GIS:

1. Complete a small digital-skills task for Arc GIS: query a map using selections (attribute and location). Create a map layout and insert map elements. Use advanced symbology and labeling. Print, export and send a map. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Arc GIS: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Arc GIS.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Arc GIS: Outcome Check

Outcome check for Arc GIS: Query a map using selections (attribute and location). Create a map layout and insert map elements. Use advanced symbology and labeling. Print, export and send a map.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Charts and objects in spreadsheet: Lesson Opener

Learning focus: Charts and objects in spreadsheet

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Charts and objects in spreadsheet.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- use charts and objects in a spreadsheet, use different techniques to organise a printable datasheet

Inquiry questions:

- How can charts and objects in spreadsheet help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Charts and objects in spreadsheet: Learn

Charts and objects in spreadsheet is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Charts and objects in spreadsheet
- Charts
- objects
- spreadsheet
- idea
- skill
- example
- practice

Charts and objects in spreadsheet: Worked Example

Worked example for Charts and objects in spreadsheet: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called charts-and-objects-in-spreadsheet-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Charts and objects in spreadsheet: Vocabulary And Study Skill

Use these words and study moves before you practise Charts and objects in spreadsheet. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Charts and objects in spreadsheet
- Charts
- objects
- spreadsheet
- idea
- skill
- example
- practice

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Charts and objects in spreadsheet without a clear example, method, or evidence.

Charts and objects in spreadsheet: Guided Activity

Guided digital activity for Charts and objects in spreadsheet:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Charts and objects in spreadsheet: Practice And Correction

Digital practice for Charts and objects in spreadsheet:

1. Complete a small digital-skills task for Charts and objects in spreadsheet: use charts and objects in a spreadsheet, use different techniques to organise a printable datasheet. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Charts and objects in spreadsheet: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Charts and objects in spreadsheet.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Charts and objects in spreadsheet: Outcome Check

Outcome check for Charts and objects in spreadsheet: Use charts and objects in a spreadsheet, use different techniques to organise a printable datasheet.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Presentation: Lesson Opener

Learning focus: Presentation

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Presentation.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- prepare and make a presentation

Inquiry questions:

- How can presentation help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Presentation: Learn

Presentation is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Presentation
- idea
- skill
- example
- practice
- correction

Presentation: Worked Example

Worked example for Presentation: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called presentation-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Presentation: Vocabulary And Study Skill

Use these words and study moves before you practise Presentation. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Presentation
- idea
- skill
- example
- practice
- correction

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Presentation without a clear example, method, or evidence.

Presentation: Guided Activity

Guided digital activity for Presentation:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Presentation: Practice And Correction

Digital practice for Presentation:

1. Complete a small digital-skills task for Presentation: prepare and make a presentation. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Presentation: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Presentation.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Presentation: Outcome Check

Outcome check for Presentation: Prepare and make a presentation.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Network security: Lesson Opener

Learning focus: Network security

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Network security.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- explain network security and apply basic security measures

Inquiry questions:

- How can network security help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Network security: Learn

Network security is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Network security
- Network
- security
- idea
- skill
- example
- practice
- correction

Network security: Worked Example

Worked example for Network security: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called network-security-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Network security: Vocabulary And Study Skill

Use these words and study moves before you practise Network security. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Network security
- Network
- security
- idea
- skill
- example
- practice
- correction

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Network security without a clear example, method, or evidence.

Network security: Guided Activity

Guided digital activity for Network security:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Network security: Practice And Correction

Digital practice for Network security:

1. Complete a small digital-skills task for Network security: explain network security and apply basic security measures. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Network security: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Network security.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Network security: Outcome Check

Outcome check for Network security: Explain network security and apply basic security measures.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Platform game programming: Lesson Opener

Learning focus: Platform game programming

Country link: a Rwandan school computer, phone, file, document, spreadsheet, network, or online-safety task for Platform game programming.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

By the end of this topic, you should be able to:

- use complex expressions, operators and controls to design platforms

Inquiry questions:

- How can platform game programming help you solve a real problem in ICT?

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Platform game programming: Learn

Platform game programming is a practical digital skill. Learn it by naming the task, following exact steps, checking the output, and using devices safely.

Core idea:

- Start with the purpose: communication, data handling, file management, research, creation, or problem solving.
- Name the tool: device, keyboard, folder, document, spreadsheet, browser, network, software, or file.
- Follow steps in order and avoid random clicking.
- Protect privacy, passwords, shared devices, and other people's files.
- Check the result by reopening, testing, proofreading, or asking whether the output matches the task.

Key words:

- Platform game programming
- Platform
- game
- programming
- idea
- skill
- example
- practice

Platform game programming: Worked Example

Worked example for Platform game programming: Organize a class assignment file safely.

1. Task: create a folder named class-work on the school computer or approved device.
2. Inside the folder, create a document called platform-game-programming-notes.
3. Type a clear heading and three short notes about the topic.
4. Save the file, close it, reopen it, and check that the content is still there.
5. Apply one safety rule: use only your own file, keep passwords private, and ask before using shared devices.

Answer check: the folder name is clear, the file opens correctly, and the learner can explain every step without guessing.

Platform game programming: Vocabulary And Study Skill

Use these words and study moves before you practise Platform game programming. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Platform game programming
- Platform
- game
- programming
- idea
- skill
- example
- practice

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Platform game programming without a clear example, method, or evidence.

Platform game programming: Guided Activity

Guided digital activity for Platform game programming:

1. Read the task and name the exact output: file, table, document, diagram, message, algorithm, or safety checklist.
2. List the device, software, data, and safety rule needed before starting.
3. Complete the output in small steps and say each step aloud to a partner.
4. Save, reopen, test, proofread, or inspect the output.
5. Write one correction you made and one rule that protects privacy or shared equipment.

Teacher check: the learner should explain the process, not only show a finished screen.

Platform game programming: Practice And Correction

Digital practice for Platform game programming:

1. Complete a small digital-skills task for Platform game programming: use complex expressions, operators and controls to design platforms. Write the exact steps, safety rule, file or device choice, check result, and one correction.

Quality checklist:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Evidence to show:

- a clear file name, table, document, diagram, algorithm, screenshot description, or step list
- one safety or privacy rule
- one test or correction

Home link: describe one safe digital habit someone at home or school should follow.

Platform game programming: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Platform game programming.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Platform game programming: Outcome Check

Outcome check for Platform game programming: Use complex expressions, operators and controls to design platforms.

1. State the digital task and the intended output.
2. List the exact device, software, file, data, or network step needed.
3. Create or describe the output in a way another learner can inspect.
4. Test the output and write one correction.
5. Add one privacy, password, copyright, ergonomics, or shared-device safety rule.

Success criteria:

- steps are exact
- digital safety is included
- output is saved, checked, or corrected

Application and utility software installation: Review Clinic 1

Use this review clinic to strengthen ICT without copying earlier answers.

1. Choose one idea from Application and utility software installation that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Charts and advanced objects: Review Clinic 2

Use this review clinic to strengthen ICT without copying earlier answers.

1. Choose one idea from Charts and advanced objects that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Arc GIS: Review Clinic 3

Use this review clinic to strengthen ICT without copying earlier answers.

1. Choose one idea from Arc GIS that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Charts and objects in spreadsheet: Review Clinic 4

Use this review clinic to strengthen ICT without copying earlier answers.

1. Choose one idea from Charts and objects in spreadsheet that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Application and utility software installation: Application Workshop 1

Digital application workshop 1 for Application and utility software installation.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Charts and advanced objects: Application Workshop 2

Digital application workshop 2 for Charts and advanced objects.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Arc GIS: Application Workshop 3

Digital application workshop 3 for Arc GIS.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Charts and objects in spreadsheet: Application Workshop 4

Digital application workshop 4 for Charts and objects in spreadsheet.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Presentation: Application Workshop 5

Digital application workshop 5 for Presentation.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Network security: Application Workshop 6

Digital application workshop 6 for Network security.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Platform game programming: Application Workshop 7

Digital application workshop 7 for Platform game programming.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Application and utility software installation: Application Workshop 8

Digital application workshop 8 for Application and utility software installation.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Charts and advanced objects: Application Workshop 9

Digital application workshop 9 for Charts and advanced objects.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Arc GIS: Application Workshop 10

Digital application workshop 10 for Arc GIS.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Charts and objects in spreadsheet: Application Workshop 11

Digital application workshop 11 for Charts and objects in spreadsheet.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Presentation: Application Workshop 12

Digital application workshop 12 for Presentation.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Network security: Application Workshop 13

Digital application workshop 13 for Network security.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Platform game programming: Application Workshop 14

Digital application workshop 14 for Platform game programming.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Application and utility software installation: Application Workshop 15

Digital application workshop 15 for Application and utility software installation.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Charts and advanced objects: Application Workshop 16

Digital application workshop 16 for Charts and advanced objects.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Arc GIS: Application Workshop 17

Digital application workshop 17 for Arc GIS.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Charts and objects in spreadsheet: Application Workshop 18

Digital application workshop 18 for Charts and objects in spreadsheet.

Grade move: apply ideas to unfamiliar situations and justify decisions with evidence; produce a checkable digital output and explain safe steps.

Use this page to make the digital skill practical and checkable.

1. Define the task and output: document, table, folder, message, search result, algorithm, diagram, or safety checklist.
2. List the device, software, file name, data, and safety rule needed.
3. Write the exact steps before using the device.
4. Create or describe the output clearly enough for another learner to inspect.
5. Test the result by reopening, checking, proofreading, tracing, or comparing with the task.
6. Record one correction and one digital-safety habit.

Extension: explain how the same skill could help a class project.

Glossary

Use this glossary to revise important words in ICT.

- Application and utility software installation: Explain this term using an example from the book, your school, or your community.
- Charts and advanced objects: Explain this term using an example from the book, your school, or your community.
- Arc GIS: Explain this term using an example from the book, your school, or your community.
- Charts and objects in spreadsheet: Explain this term using an example from the book, your school, or your community.
- Presentation: Explain this term using an example from the book, your school, or your community.
- Network security: Explain this term using an example from the book, your school, or your community.
- Platform game programming: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong ICT answer responds to the exact task, uses correct vocabulary, gives a clear example or evidence, and shows correction after feedback. For practical work, name materials, safety, process, quality checks, peer feedback, and one improvement made after review.

Final Project

Create a final project for ICT that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.